Navid Safahieh

Department of Economics, Mitch Daniels School of Business
Purdue University, 403 Mitch Daniels Blvd, West Lafayette, IN 47907
+1 (765) 771–0021 · nsafahie@purdue.edu · navidzky.github.io
Nationality: New Zealand / Iran

On the job market 2026–27

Research Interests

Transportation Economics, Search and Matching, Macroeconomics, Causal Inference

Education

Ph.D. in Economics Expected 2027
Purdue University, Mitch Daniels School of Business, West Lafayette, IN

M.Sc. in Economics 2021
University of Tehran, Tehran, Iran

B.Sc. in Civil and Environmental Engineering 2018
Sharif University of Technology, Tehran, Iran

Dissertation Committee: Cathy Zhang (Chair), Mario J. Crucini, Chong Xiang

Job Market Paper

Match Now, Arrive Later: Transport Markets with Remote Search

In transport markets, carriers are often hired while still far from the cargo, and the customer waits while the carrier travels toward the cargo. I develop a dynamic spatial directed search model that allows for this remote matching. In existing models of transport markets, matching is local: a carrier must sail to a cargo before it can compete for it, sinking the cost of the trip for a cargo it may not win. With remote matching, a carrier competes from where it is for cargoes at every location, and a customer can attract carriers from every location, at the cost of waiting while they travel, so competition for carriers and for cargoes extends across locations. In a two-location example in which cargo moves in one direction only, as trips become more valuable, carriers stop sailing to the cargo before they are hired and wait to be hired remotely, even though being near the cargo remains more valuable: the cost of the time a carrier spends sailing without a contract, instead of applying, grows faster with the value of a trip than the premium of being near the cargo. Customers then hire more distant carriers and wait longer, consistent with tankers being fixed farther from the loading port when freight markets are strong. Compared with an otherwise identical model with local matching only, once trips are valuable enough, carriers are worth more at every location and the premium of being near the cargo is smaller. In an application with the Persian Gulf, Europe and Asia, I close Bab el-Mandeb, which lengthens voyages from the Gulf to Europe. The outcome depends on the value of a trip to Europe. When it is high enough that the detour is worth taking, the shock negatively affects the route to Asia, which does not use the passage: its fee rises and less cargo moves on it. When it is lower, carriers shift to the Asia route and its fee falls. Over a wide range of values, remote matching amplifies the spillover: customers shipping to Europe keep shipping by hiring distant carriers, and shipments to Asia fall more than with local matching.

Publications

Modeling the Accuracy of Traffic Crash Prediction Models

- *IATSS Research*, 2022. With Mohammad Hesam Rashidi, Soheil Keshavarz, Parham Pazari, and Amir Samimi.
- Models forecasting accuracy across 31 Iranian provinces using macroeconomic variables and road traffic indicators. Crash variability, average historical crash count, employment ratio, GDP per capita, and traffic exposure are the variables that affect forecast accuracy.

Working Papers

When Outside Options Shape the Labor Market: A Directed Search Perspective

- Studies how differences in outside options across workers affect wages, firm entry, and job allocation. I develop a directed search model with skilled and unskilled workers and high- and low-tech firms, where only skilled workers hold outside options. Stronger outside options raise posted wages for skilled workers but can reduce high-tech firm entry; as fewer high-tech jobs are created, skilled workers' expected wages may fall, and unskilled workers are affected indirectly through changes in firm composition and wage competition.

The Heterogeneous Effects of Exogenous Shocks on Public Transportation Demand

- With Kowsar Yousefi and Amir Samimi.
- A sudden gasoline price shock increased subway usage by roughly 3.9 percent, implying a cross-price elasticity of about 0.044, while subway usage fell by roughly 42.6 percent during the COVID-19 pandemic.

Conference Presentations

- MEA 90th Annual Meetings, Chicago, IL — March 2026

Additional Training

- Estimation of Structural Models with Conditional Choice Probabilities, Notre Dame, IN — Summer 2025
- 14th Annual Research Design for Causal Inference, Northwestern University, IL — Summer 2025

Teaching Experience

ECON 252 Principles of Macroeconomics (Undergraduate) — Fall 2025
Instructor — Purdue University

ECON 252 Principles of Macroeconomics (Undergraduate) — Summer 2025
Instructor — Purdue University

Intermediate Macroeconomics (Undergraduate) — Fall 2025
Teaching Assistant to Prof. Cathy Zhang — Purdue University

Macroeconomics (PhD) — Fall 2023
Teaching Assistant to Prof. Seunghoon Na — Purdue University

Econometrics (Graduate) — 2019 – 2021
TA to Profs. Kowsar Yousefi, Atiyeh Vahidmanesh, and Alireza Oryoie — University of Tehran

Game Theory (Graduate) — Fall 2019
Teaching Assistant to Prof. Ghahreman Abdoli — University of Tehran

Econometrics in Transportation (Graduate) — 2020 – 2021
Teaching Assistant to Prof. Amir Samimi — Sharif University of Technology

Fellowships and Awards

- Bilsland Dissertation Fellowship, Purdue University — 2026–27
- Outstanding Teaching Award (ECON 252), Purdue University — Fall 2025
- Distinguished Teaching Award (ECON 252), Purdue University — Summer 2025

Service

- Economics Senator, Business Doctoral Students Association, Purdue University

Programming and Data Skills

Python, R, STATA, MATLAB, Claude Code

Structural modeling, numerical equilibrium computation, simulation-based methods.

Languages

Farsi (Native), English (Fluent)

References

Cathy Zhang (Chair)
Professor
Mitch Daniels School of Business
Purdue University
cmzhang@purdue.edu

Mario J. Crucini
Professor
Mitch Daniels School of Business
Purdue University
mcrucini@purdue.edu

Chong Xiang
Professor
Mitch Daniels School of Business
Purdue University
cxiang@purdue.edu